

Lesson Three Overview



Throughout the lecture, we are giving you some **“food for thought”**, these are questions we suggest you think about as you go through this lesson. You do not need to submit solutions.

Learning Objectives

Students will be able to:

- Measure and interpret the degree distribution of a network
- Understand the *“friendship paradox”* to illustrate the importance of the degree distribution
- An application of the friendship paradox: vaccination targets when the network topology is unknown
- Learn the $G(n,p)$ model as the most basic type of random graph
- Degree correlations and assortative networks

Required and Recommended Reading

Required Reading

- Chapter-3 from A-L. Barabási, [Network Science](http://www.networksciencebook.com/) (<http://www.networksciencebook.com/>), 2015.
- Chapter-7 (sections 7.1, 7.2, 7.3, 7.5) from A-L. Barabási, [Network Science](http://www.networksciencebook.com/) (<http://www.networksciencebook.com/>), 2015.

Recommended Reading

- [Simulated Epidemics in an Empirical Spatiotemporal Network of 50,185 Sexual Contacts](http://www.ploscompbiol.org/article/info%3Adoi%2F10.1371%2Fjournal.pcbi.1001109#s2).  (<http://www.ploscompbiol.org/article/info%3Adoi%2F10.1371%2Fjournal.pcbi.1001109#s2>) Luis E. C. Rocha, Fredrik Liljeros, Petter, Holme (2011)

Assignments

 L3: Knowledge Check

L3: Quiz